

Direct Connect

Dedicated, Private Physical Connectivity Between On-Prem and AWS

Interview Preparation Guide

Problem & motivation - VIFs & BGP - resiliency models - packet flow - failure modes - scaling - security - observability & automation - cost - troubleshooting - real-world patterns

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Executive Summary

AWS Direct Connect (DX) provides a dedicated, private physical network connection between an on-premises data center and AWS, bypassing the public internet. It delivers consistent bandwidth, lower and more predictable latency, and reduced data-transfer-out costs at scale. Logical Virtual Interfaces (private, public, or transit) ride over the physical link, exchanging routes via BGP. Because DX is physical, it has long lead times and must be paired with redundancy (a second DX or a VPN backup) for production. This guide covers internals, packet flow, failure modes, scaling, security, observability, automation, cost, and troubleshooting.

1. The Problem It Solves

Internet-based VPN connectivity to AWS rides shared, best-effort internet paths with variable latency, jitter, and throughput, and incurs internet data-transfer charges. Workloads that need consistent performance, high sustained bandwidth, or lower egress cost need a dedicated path - that is Direct Connect.

Core needs DX addresses

- **Consistent performance:** Dedicated bandwidth with predictable, low latency - not best-effort internet.
- **High throughput:** 1/10/100 Gbps dedicated, or sub-1G hosted connections.
- **Lower egress cost:** Reduced data-transfer-out rates vs internet at scale.
- **Private reachability:** Reach VPCs and AWS public services without traversing the public internet.

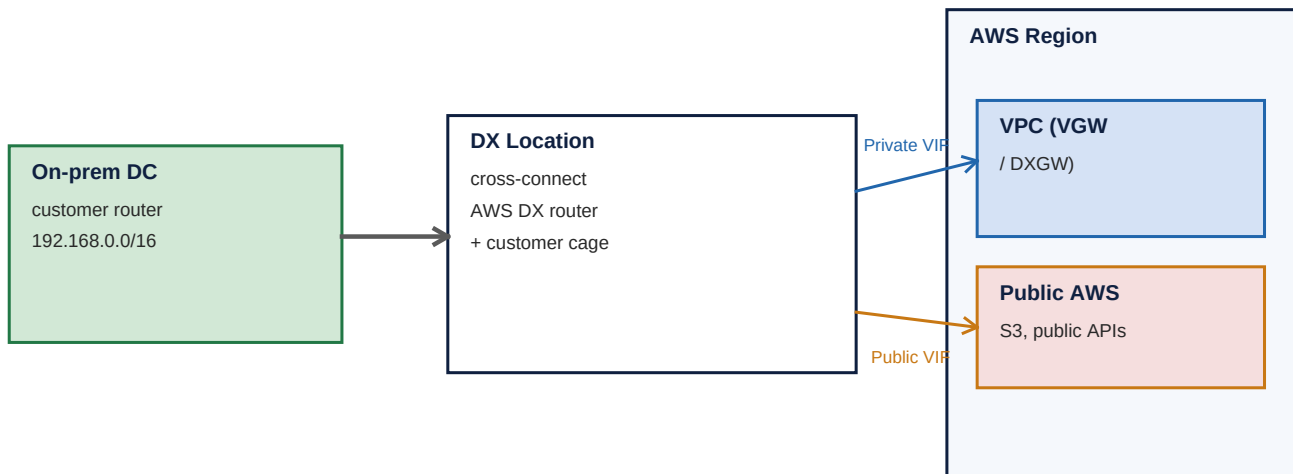
2. How It Works Internally

You establish a cross-connect at a Direct Connect location between your router (in a colo cage or via a partner) and an AWS DX router. Over that physical port you create Virtual Interfaces (VIFs): a PRIVATE VIF to reach VPCs (through a Virtual Private Gateway or a Direct Connect Gateway), a PUBLIC VIF to reach AWS public services (e.g., S3) over the private link, or a TRANSIT VIF to reach a Transit Gateway. BGP sessions exchange routes between your network and AWS.

- Dedicated connection: a physical AWS port (1/10/100 Gbps). Hosted connection: capacity provisioned via a partner (often sub-1G).
- A Direct Connect Gateway lets one VIF reach VPCs in multiple regions/accounts.
- BGP advertises your on-prem prefixes to AWS and AWS/VPC prefixes to you.
- DX is NOT encrypted by itself - layer IPsec/MACsec for confidentiality.

Reference Diagram - Direct Connect private link to AWS

AWS Direct Connect (DX) is a dedicated, private physical network link between your data center and AWS, bypassing the public internet for consistent bandwidth and lower, more predictable latency. Logical Virtual Interfaces (VIFs) ride the physical link.



A private VIF reaches VPCs (via a Virtual Private Gateway or Direct Connect Gateway); a public VIF reaches AWS public services; a transit VIF reaches a Transit Gateway. BGP exchanges routes. Provisioning takes weeks (physical cross-connect), so a VPN backup is standard.

3. Traffic Flow, Packet Flow & Routing Decisions

On-prem to VPC over a private VIF

- On-prem host -> customer router; BGP has learned the VPC CIDR via DX.
- Traffic crosses the cross-connect to the AWS DX router over the private VIF.
- Routed via VGW/DX Gateway into the target VPC; VPC route tables carry the on-prem prefix back via the gateway.
- Security groups/NACLs apply inside the VPC as usual.
- Return traffic follows BGP-learned routes back over DX.

Routing preference & resiliency

When both DX and a VPN exist, BGP normally prefers DX; the VPN takes over on DX failure. Tune BGP (AS-path prepend, local pref, MED) for deterministic primary/backup behavior.

4. Advantages & Disadvantages

Advantages

- Consistent, low-latency, high-bandwidth private connectivity.
- Lower data-transfer-out cost than internet at scale.
- Reaches VPCs (multi-region via DX Gateway) and AWS public services.
- Avoids the public internet entirely for the transport.

Disadvantages / costs

- Long lead time (physical cross-connect) - provisioning takes weeks.
- Single DX is a SPOF; true resiliency needs redundant links/locations.
- Not encrypted by default - add IPsec/MACsec.
- Port-hour + data-transfer charges, plus colo/partner costs.

5. Common Failure Modes

Failure	Symptom	Typical root cause
Single DX SPOF	Total hybrid outage	No redundant connection or VPN backup.
BGP misconfig	No/partial routes	Wrong ASN, prefixes not advertised, or filters.
No encryption	Compliance gap	Assuming DX is private == encrypted.
MTU mismatch	Fragmentation/drops	Jumbo-frame settings inconsistent.
VIF/gateway gap	VPC unreachable	Private VIF not associated with VGW/DXGW correctly.
Asymmetric routing	One-way	DX in, VPN out (or vice versa) due to route preference.

6. Scaling Considerations

- **Redundancy:** Two connections at two DX locations (and/or VPN backup) for production SLAs.
- **LAG:** Aggregate multiple ports into a Link Aggregation Group for more bandwidth.
- **DX Gateway:** Reach many VPCs across regions/accounts from one connection.
- **Transit VIF + TGW:** Scale to many VPCs via a Transit Gateway instead of per-VPC private VIFs.
- **Capacity planning:** Choose dedicated (1/10/100G) vs hosted based on sustained throughput.

7. Security Implications

- DX is private transport but NOT encrypted - add IPsec VPN over DX or MACsec.
- Filter BGP advertisements to avoid leaking/accepting unintended routes.
- Apply SGs/NACLs inside the VPC; DX does not filter traffic.
- Protect the physical/colo footprint and partner relationships.

8. Observability, Reliability & Automation

Observability

- CloudWatch DX metrics: ConnectionState, BGP state, light levels, in/out bps, error counts.
- Alarm on BGP session down and link/light degradation.

Reliability

- Design to AWS's resiliency models (redundant connections/locations).
- Maintain a VPN failover and test it; tune BGP for clean failover.

Automation

- Manage VIFs, DX Gateway, and associations in IaC where supported.
- Automated BGP/route monitoring and failover tests.

9. Cost Considerations

Cost driver	Why it adds up	Mitigation
Port-hours	Charged per hour for the connection capacity.	Right-size capacity; use hosted for smaller needs.
Data transfer out	Lower DX rate, but still per-GB.	Route bulk egress over DX vs internet at scale.
Redundancy	Second link/location multiplies cost.	Balance SLA vs cost; VPN backup is cheaper than dual DX.

10. Troubleshooting Approach

For 'on-prem cannot reach the VPC over DX':

- 1. Physical/port: is the connection 'up' and light levels healthy?
- 2. BGP: are sessions established and the right prefixes advertised/received?
- 3. VIF: correct type and associated with VGW/DXGW/TGW?
- 4. VPC routes: do subnet tables carry the on-prem prefix via the gateway?
- 5. Return path & preference: is routing symmetric (DX vs VPN)? Check MTU.

Mnemonic

'Port -> BGP -> VIF -> VPC routes -> Symmetry.' Most DX issues are BGP advertisement/preference or a missing VIF-to-gateway association.

11. Real-World Examples & Patterns

- **Dual DX:** Two connections at two locations for high resiliency.
- **DX + VPN backup:** DX primary with an IPsec VPN failover (cost-effective resiliency).
- **Encrypted DX:** IPsec/MACsec over DX for confidentiality + compliance.
- **DX Gateway + TGW:** One link reaching many VPCs/regions via transit VIF + Transit Gateway.

12. Senior-Level Trade-offs & Talking Points

- DX (consistent, private, costly, slow to provision) vs VPN (fast, cheap, internet-variable).
- Dedicated vs hosted connections: capacity/control vs speed-to-provision.
- Dual DX (highest SLA) vs DX + VPN backup (cost-effective resiliency).
- Encryption overhead vs compliance requirements over a 'private' link.

Whiteboard checklist

(1) Draw on-prem -> DX location cross-connect -> AWS. (2) Show private/public/transit VIFs. (3) BGP route exchange. (4) Add VPN backup + BGP preference. (5) Note no default encryption. (6) Discuss resiliency models and cost.

13. Quick Glossary

Term	Meaning
Direct Connect	Dedicated private physical link from on-prem to AWS.
VIF	Virtual Interface (private/public/transit) over the DX link.
Private VIF	Reaches VPCs via a VGW or Direct Connect Gateway.
Public VIF	Reaches AWS public services over the private link.
DX Gateway	Connects a VIF to VPCs across regions/accounts.
LAG	Link Aggregation Group bundling multiple ports.
BGP	Routing protocol exchanging prefixes between on-prem and AWS.